Health Probes Lab

Liveness · Readiness · Startup Probe · All 3 Methods · Debug

⏱ 50 minutes 📊 100 points 📋 5 tasks 🎯 CKA Level

Session: S12

💗 Part 1 — Liveness Probe

30 pts

1 Liveness Probe — HTTP GET Method

Easy ⭐

Liveness probe tells K8s: is this container still alive? If it fails → restart the container.

Field	Value
Pod name	`liveness-http`
Image	`nginx:latest`
Probe type	`httpGet`
Path	`/`
Port	`80`
initialDelaySeconds	`5`
periodSeconds	`10`
failureThreshold	`3`

```
    livenessProbe:
      httpGet:
        path: /
        port: 80
      initialDelaySeconds: 5   # wait 5s before first check
      periodSeconds: 10        # check every 10s
      failureThreshold: 3      # restart after 3 consecutive failures
```

Requirements:

a Apply the pod with liveness probe

b Confirm the probe is configured

```
kubectl describe pod liveness-http | grep -A8 Liveness
```

c Simulate failure by breaking nginx

```
kubectl exec liveness-http -- rm /usr/share/nginx/html/index.html
```

```
kubectl get pod liveness-http -w
```

✍️ What happened to the pod after the liveness probe failed 3 times?

The pod was restarted

✍️ What does the RESTARTS column show?

The number of times the pod has been restarted (1)

2 Liveness Probe — Exec Command Method

Medium ⭐⭐

Field	Value
Pod name	`liveness-exec`
Image	`busybox`
Probe type	`exec`
Command	`cat /tmp/healthy`
Strategy	`file exists = healthy, file deleted = probe fails`

```
spec:
  containers:
  - name: app
    image: busybox
    command: ['sh','-c','touch /tmp/healthy; sleep 30; rm /tmp/healthy; sleep 600']
    livenessProbe:
      exec:
        command: ['cat', '/tmp/healthy']
      initialDelaySeconds: 5
      periodSeconds: 5
      failureThreshold: 2
```

Requirements:

- **a** Apply the pod — it creates /tmp/healthy on start
- **b** Wait 35 seconds — container deletes /tmp/healthy on its own
- **c** Watch the restart happen automatically

```
kubectl get pod liveness-exec -w
```

✍️ How long did it take for the restart to happen after the file was deleted?

10 ~ 15 seconds after the file was deleted

✍️ Formula: time-to-restart = initialDelaySeconds + (failureThreshold × periodSeconds). Calculate it.

5 + (2 × 5) = 15 seconds

🚦 Part 2 — Readiness Probe 35 pts

3 Readiness Probe — Remove from Service When Not Ready Medium ⭐⭐

Readiness probe tells K8s: is this container ready to receive traffic? If it fails → remove from Service endpoints (do NOT restart).

Field	Value
Deployment name	`ready-demo`
Replicas	`2`
Image	`nginx:latest`
Readiness path	`/ on port 80`
Test	`break nginx on one pod — see it removed from endpoints`

```
    readinessProbe:
      httpGet:
        path: /
        port: 80
      initialDelaySeconds: 3
      periodSeconds: 5
      failureThreshold: 2
```

Requirements:

a Create deployment with readiness probe and expose as service

```
kubectl apply -f ready-demo.yaml
```

```
kubectl expose deployment ready-demo --port=80
```

b Check endpoints — both pod IPs should be there

```
kubectl get endpoints ready-demo
```

c Break nginx on ONE pod

```
kubectl exec <pod-name-0> -- rm /usr/share/nginx/html/index.html
```

d Check endpoints again — unhealthy pod should be removed

```
kubectl get endpoints ready-demo
```

```
kubectl get pods
```

✍️ Was the unhealthy pod RESTARTED or just REMOVED from endpoints?

Removed from endpoints

✍️ Key difference between Liveness and Readiness probe — when to use each:

Liveness restarts unhealthy pods
Readiness stops the traffic until the pod is ready.

4 Startup Probe — Protect Slow-Starting Apps

Medium ⭐⭐

Startup probe gives slow apps time to start before liveness kicks in. Prevents premature restarts.

Field	Value
Pod name	`slow-app`
Startup probe	`HTTP GET / — failureThreshold 30 × periodSeconds 10 = 300s to start`
Liveness probe	`same path — only activates AFTER startup probe succeeds`
Formula	`max startup time = failureThreshold × periodSeconds`

```
    startupProbe:
      httpGet:
        path: /
        port: 80
      failureThreshold: 30    # allow up to 300 seconds to start
      periodSeconds: 10
    livenessProbe:
      httpGet:
        path: /
        port: 80
      periodSeconds: 10
      failureThreshold: 3    # only runs AFTER startup probe passes
```

Requirements:

- **a** Apply the pod
- **b** Describe the pod — confirm both probes are configured

```
kubectl describe pod slow-app | grep -A6 -E 'Liveness|Startup'
```

✍️ If failureThreshold=30 and periodSeconds=10 — how long does the app have to start?

300 seconds (5 minutes).

✍️ What happens to liveness probe while startup probe is still running?

It is disabled and does not run

🔧 Part 3 — All Three Together + Debugging 35 pts

5 Production Pod — All 3 Probes Combined Hard ⭐⭐⭐

A production-ready pod uses all 3 probes together. This is the pattern to know for CKA.

```
spec:
  containers:
  - name: app
    image: nginx:latest
    ports:
    - containerPort: 80
    startupProbe:
      httpGet: { path: /, port: 80 }
      failureThreshold: 10
      periodSeconds: 5            # up to 50s to start
    readinessProbe:
      httpGet: { path: /, port: 80 }
      initialDelaySeconds: 0
      periodSeconds: 5
      failureThreshold: 2         # 10s to remove from Service
    livenessProbe:
      httpGet: { path: /, port: 80 }
      periodSeconds: 10
      failureThreshold: 3         # 30s of failure before restart
```

Requirements:

a Apply this production pod

b Confirm all 3 probes show in describe

```
kubectl describe pod production-app | grep -E 'Liveness|Readiness|Startup' -A8
```

c Verify pod is Ready (readiness passed)

```
kubectl get pod production-app
```

d Test the probe ordering by watching events

```
kubectl describe pod production-app | grep -A20 Events
```

✅ **Verify:**

```
Pod status: Running
READY: 1/1 (readiness probe passed)
RESTARTS: 0 (liveness probe happy)
```

✍️ In what ORDER do the 3 probes run? (Which one first?)

Startup -> Readiness -> Liveness Startup must pass first

✍️ What would you change to give an app 5 minutes to start?

Set failureThreshold: 30 and periodSeconds: 10